

Tree Fruit Guidelines

2026-07-10

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1 Tree Fruit Guideline

- 1.1 Test
- 1.2 Macmuncher applemunchus voracious

2 1.1 Test

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2.1 Header 2

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1. List Item 1
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3 1.2 Mac Muncher (Applemunchus voracious)

The Mac Muncher (*Applemunchus voracious*) is a stout, apple-loving beetle famous for its seemingly endless appetite and complete inability to finish what it starts. Adults typically measure 8–12 mm in length and possess a glossy olive-green shell decorated with pale cream spots resembling apple blossoms. Their powerful reddish mandibles are capable of chewing through even the crispest apple skin.

While technically capable of eating other fruits, Mac Munchers stubbornly refuse to do so. Scientists have repeatedly offered them pears, peaches, and cherries, only to discover the insects sitting nearby in apparent protest.

Tree Fruit 6.4.1 Activity Spectrum of stone fruit fungicides.

Fungicide1	Fungicide Family	FRAC Code3	registered for use on:	Header 8	Header 9	Header 10	Header 11
Apircot	Cherry	Peach	Plum				
Aprovia							

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Pesticide Summary Table: 1.1 Pesticide Summary Table Highbush Blueberries, Insecticides

The species is most often found in orchards, backyard apple trees, and anywhere an unattended pie is cooling on a windowsill.

1.1 Mac Muncher, on Apples

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
* Assail 30SC (acetamiprid)	18 gal / acre	1 hr; Until Dry	2 Days; Refer To Label	IRAC: 4A	Moderate
Assail 30SC has demonstrated good efficacy against Mac Munchers (<i>Applemunchus voracious</i>). Growers may notice a significant reduction in fruit damage and a corresponding decrease in tiny beetles acting like self-appointed orchard taste testers.					
* CrunchStop Max (Anti-Crunchate)	10 gal / acre	12 hr; Refer To Label	1 Days; Until Dry		High
CrunchMax delivers excellent knockdown of Mac Muncher populations. Field observations indicate a significant reduction in the number of apples receiving the customary "quality assurance bite."					

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
The Orchard Inspector Beetle: The Orchard Inspector Beetle is one of the Mac Muncher's most feared natural enemies. Roughly twice the size of an adult Mac Muncher, this shiny blue-black beetle patrols orchards searching for apples that have been partially eaten. Unlike most predators, the Orchard Inspector does not immediately attack. Instead, it carefully examines the damage, takes notes, shakes its head in disappointment, and then tracks the guilty Mac Muncher through the orchard. Researchers believe Mac Munchers can somehow sense when an Orchard Inspector is nearby. Upon detection, many immediately abandon their current apple and attempt to blend in by pretending to be seeds, leaves, or decorative garden ornaments. The Orchard Inspector Beetle is one of the orchard's most dedicated defenders, actively hunting Mac Munchers throughout the growing season while leaving fruit unharmed. A single beetle may consume hundreds of Mac Munchers per year, making it a valuable biological control agent. However, growers should not be alarmed if these beetles are observed inspecting perfectly healthy apples, taking lengthy notes, or appearing mildly disappointed with the orchard's overall management. Their standards are notoriously high.					

3.1 1.2.1 Codling Moth

The Codling Moth (*Cydia pomonella*) is a primary pest of apples and pears, with larvae causing “wormy” fruit by burrowing into the core to feed on seeds. Cornell IPM materials indicate that this species completes multiple generations annually, with larvae creating destructive, unmarketable tunnels. For more details, visit Cornell IPM.

1.2 Codling Moth, on Apples

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
Checkmate CM 2.0 (pheromone)	2.4-4.8 fl oz / acre	4 hr	Refer To Label		Moderate

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations.					
Isomate-CM/OFM TT (pheromone)	200 ties/ / acre	0 hr	Refer To Label		High
Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations.					
* Baythroid XL (beta-cyfluthrin)	2-2.4 fl oz / acre	12 hr	7 Days	IRAC: 3A	Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
Cyd-X 0.06SC (granulosis virus)	1-6 fl oz / acre	4 hr	0 Days		Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. Cyd-X labeled for use against codling moth only.					
Deliver (Bacillus thuringiensis, subsp. kurstaki)	0.5-2 lb / acre	4 hr	0 Days		Moderate

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
Delegate WG (spinetoram)	4.5-7 oz / acre	4 hr	7 Days	IRAC: 5	High
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
Dipel 10.3DF (Bt)	0.5-2 lb / acre	4 hr	0 Days		Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. Dipel and *Imidan not registered for lesser appleworm.					
Entrust 2SC (spinosad)	6-10 fl oz / acre	4 hr	7 Hours; Refer To Label	IRAC: 5	Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
Entrust 80WP (spinosad)	0.67-1 oz / 100 gal water; 2-3 oz / acre	4 hr	7 Days	IRAC: 5	Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
* Imidan 70-W (phosmet)	0.75-1.0 lb / 100 gal water; 2.13-5.75 lb / acre	(E) hr	7 Days	IRAC: 1B	High
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. Dipel and *Imidan not registered for lesser appleworm.					
* Intrepid 2F (methoxyfenozide)	12-16 fl oz / acre	4 hr	14 Days		Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. *Altacor and *Intrepid provide only suppression of codling moth; not registered for use in Nassau or Suffolk Counties.					
* Lannate LV (methomyl)	1.5 pts / 100 gal water; 1.5-3 pts / acre	72 hr; Refer To Label	14 Days	IRAC: 1A	High

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

*Lannate not registered for oriental fruit moth.

* Lannate SP (methomyl)	0.25 lb / 100 gal water; 0.5-1 lb / acre	72 hr; Refer To Label	14 Days	IRAC: 1A	High
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*Lannate not registered for oriental fruit moth.

Neemix (azadirachtin)	7-16 fl oz / acre	4 hr	0 Days	IRAC: UN	Moderate
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

Checkmate Puffer CM-OFM (pheromone)	1-2 units/ / acre	0 hr	Refer To Label		Moderate
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Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations.

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
* Proclaim (emamectin benzoate)	1.2 oz / 100 gal water; 4.8 oz / acre	12/48(E) hr	14 Days	IRAC: 6	Moderate

For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

*Proclaim used against codling moth should be applied at egg hatch and may be followed up with a second application in 7-14 days.

* Rimon 0.83EC (novaluron)	20-40 fl oz / acre	12 hr	14 Days	IRAC: 15	High
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

*Rimon is limited to 1 application per season.

Sevin XLR Plus (carbaryl)	1-3 qt / acre	12 hr	3 Days	IRAC: 1A	Moderate
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

Use Sevin for codling moth and lesser appleworm; do not use within 30 days of full bloom unless fruit thinning is desired.

Surround 95WP (kaolin)	25-50 lb / acre	4 hr	UDH Days		Moderate
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* Endigo ZC (thiamethoxam, lambda-cyhalothrin)	5-6 fl oz / acre	24 hr	35 Days	IRAC: 4A, 3A	Moderate
For best effectiveness and insecticide resistance management, the use of pre-mixes should be reserved for situations when multiple pest species are present and appropriately matched to the combination of active ingredients and modes of action contained in the product. For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. Do not exceed 0.172 lb a.i./A of thiamethoxam- containing products per acre per growing season.					
* Leverage 360 (imidacloprid, beta-cyfluthrin)	2.4-2.8 fl oz / acre	12 hr	7 Days	IRAC: 4A, 3A	High
For best effectiveness and insecticide resistance management, the use of pre-mixes should be reserved for situations when multiple pest species are present and appropriately matched to the combination of active ingredients and modes of action contained in the product. For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
* Voliam Flexi WDG (thiamethoxam, chlorantraniliprole)	4-7 oz / acre	12 hr	35 Days; Refer To Label	IRAC: 4A, 28	High

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

Do not exceed 0.172 lb a.i./A of thiamethoxam- containing products per acre per growing season.

Voliam Xpress EC (Lambda-cyhalothrin/chlorantraniliprole)	6-12 fl oz / acre			IRAC: 3A	High
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

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*Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].

* Gladiator EC (zeta-cypermethrin, avermectin B1)	4.75 fl oz / 100 gal water; 19 fl oz / acre	12 hr	28 Days	IRAC: 3A, 6	Moderate
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For best effectiveness and insecticide resistance management, the use of pre-mixes should be reserved for situations when multiple pest species are present and appropriately matched to the combination of active ingredients and modes of action contained in the product.

Madex HP (granulosis virus)	0.5-3 fl oz / acre	4 hr	0 Days		Moderate
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For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website

(<http://newa.cornell.edu/index.php?page=apple-insects>) for current information on the occurrence, development and management of these pests in your specific location.

Sprays of Madex or Virosoft should target the 5% hatch point of each generation, and can be applied on a 7-day interval.

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
Isomate CM/OFM Mist Plus (pheromone)	1-2 units/ / acre				High
Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations. Place aerosol units in upper third of tree canopy and at least 3 feet from fruits, facing toward the drive row; distribute units in a grid pattern to achieve one or two units per acre. Best used in large contiguous areas (>40 acres) for optimal results.					
* Besiege (chlorantraniliprole, lambda-cyhalothrin)	6-12 fl oz / acre	24 hr	21 Days	IRAC: 28, 3A	High
For best effectiveness and insecticide resistance management, the use of pre-mixes should be reserved for situations when multiple pest species are present and appropriately matched to the combination of active ingredients and modes of action contained in the product. For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1].					
* Minecto Pro (abamectin, cyantraniliprole)	8.0-12.0 fl oz / acre	12 hr	28 Days	IRAC: 6, 28	High
For best effectiveness and insecticide resistance management, the use of pre-mixes should be reserved for situations when multiple pest species are present and appropriately matched to the combination of active ingredients and modes of action contained in the product. For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. Not registered for use in Nassau or Suffolk Counties.					
Cidetrak CM-OFM Combo (codling moth/oriental fruit moth pheromone)	200-220 dispensers/A				Moderate

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations.					
Virosoft CP4 (insect virus)	1.6-3.2 fl oz / acre				Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. Sprays of Madex or Virosoft should target the 5% hatch point of each generation, and can be applied on a 7-day interval.					
Cidetrak CMDA Combo Meso-A (codling moth pheromone)	18-36 dispensers / acre				Moderate
Better control of target species is obtained when pheromone disruption begins with the first generation of the season; regardless, products for disruption should be applied before first flight of the generation being targeted. Products directed against oriental fruit moth (Checkmate, Isomate, Cidetrak products) are incidentally active against lesser appleworm. CM-OFM combination products are active against the above species as well as codling moth. The need for re-application depends on residual field life; for Checkmate OFM-F and CM 2.0, 14-21 days, Cidetrak products, 120 days. Insecticide sprays or double the rate of pheromones may be needed in border rows of orchards adjacent to sources of adult immigration or in other high pressure situations.					
Spear-Lep (GS-omega/kappa-Hxtx-Hv1a)	1-2 pts / acre	4 hr	0 Days	IRAC: 32	-
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. Spear-Lep must be tank mixed with a <i>Bacillus thuringiensis</i> (Bt) product or another product that assists in moving the active ingredient to the site of activity.					
* Mustang MAXX (zeta-cypermethrin)	1.28-4.0 fl oz / acre	12 hr	14 Days	IRAC: 3A	High

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. Do not apply this product while bees are foraging; do not apply until flowering is complete and all petals have fallen. If an application must be made when managed bees are at the treatment site, the beekeeper providing the pollination services must be notified no less than 48 hours prior to the time of the planned application so that the bees can be removed, covered or otherwise protected prior to spraying.					
Avaunt 30WDG (indoxacarb)	5-6 oz / acre	12 hr	14 Days	IRAC: 22A	Moderate
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Altacor, Avaunt, *Assail, *Cyclaniliprole and *Verdepryn applied at petal fall will also control European apple sawfly. Avaunt and *Verdepryn will control plum curculio. *Verdepryn will also control codling moth and obliquebanded leafroller.					
* Exirel (cyantraniliprole)	8.5-17 fl oz / acre	12 hr	3 Days	IRAC: 28	High
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. Do not apply this product while bees are foraging; do not apply until flowering is complete and all petals have fallen. If an application must be made when managed bees are at the treatment site, the beekeeper providing the pollination services must be notified no less than 48 hours prior to the time of the planned application so that the bees can be removed, covered or otherwise protected prior to spraying. Not registered for use in Nassau or Suffolk Counties. For each generation, make the first application of *Exirel prior to egg hatch. For low pressure infestations, use the 8.5-10 fl oz rate with repeat applications on a 14-day interval. *Exirel applied at petal fall and 1st cover will also provide control of apple leafcurling midge.					
Grandevo WDG (<i>Chromobacterium subtsugae</i> str. PRAA4-1)	1-3 lb / acre	4 hr	0 Days		Moderate

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location.					
* Verdepryn 100 SL (cyclaniliprole)	5.5-11.0 fl oz / acre	4 hr	7 Days	IRAC: 28	High
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Altacor, Avaunt, *Assail, *Cyclaniliprole and *Verdepryn applied at petal fall will also control European apple sawfly. Avaunt and *Verdepryn will control plum curculio. *Verdepryn will also control codling moth and obliquebanded leafroller. Do not apply this product while bees are foraging; do not apply until flowering is complete and all petals have fallen. If an application must be made when managed bees are at the treatment site, the beekeeper providing the pollination services must be notified no less than 48 hours prior to the time of the planned application so that the bees can be removed, covered or otherwise protected prior to spraying. Not registered for use in Nassau or Suffolk Counties.					
* Altacor (chlorantraniliprole)	2.5-4.5 oz / acre	4 hr	5 Days	IRAC: 28	High
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. *Altacor, Avaunt, *Assail, *Cyclaniliprole and *Verdepryn applied at petal fall will also control European apple sawfly. Avaunt and *Verdepryn will control plum curculio. *Verdepryn will also control codling moth and obliquebanded leafroller. *Altacor and *Intrepid provide only suppression of codling moth; not registered for use in Nassau or Suffolk Counties.					
Dipel DF (Bacillus thuringiensis, subsp. kurstaki)	0.5-2 lb / acre	4 hr	0 Days		Moderate

Control Technique	Rate	REI	PHI	Resistance Mgmt.	Efficacy
For orchards not receiving regular apple maggot sprays, or where codling moth is otherwise a significant problem, a developmental model predicts the appropriate larval treatment period for CM as 250-360 degree-days (base 50F) after 1st adult catch for each generation, and approximately 150 DD after this same biofix date for insecticides with ovicidal activity. Sprays of granulosis virus should target the 5% hatch point of each generation, and can be applied on a 7-14-day interval. Refer to the NEWA Apple Insect Models website (http://newa.cornell.edu/index.php?page=apple-insects) for current information on the occurrence, development and management of these pests in your specific location. *Asana or *Warrior applied during this period will also provide control of these pests, as will a spray program as noted for apple maggot. Suggested action threshold: Avg. of >5 CM adults/week caught per pheromone trap once 150-360 DD (base 50F) have accumulated since biofix; see [14.1]. Dipel and *Imidan not registered for lesser appleworm.					

4 Test

Test

Test

Test

Test

Header 1	Header 2	Header 3	Header 4
a	1	2	
b	1b	2bc	
c	1c		
d	1d		
e	1e	2e	

